Define Problem Statements

Phase 1: Brainstorming & Ideation

1. Background

A travel plan is assembled from several decisions: where to go, when to travel, how much can be spent, who is travelling, and which activities matter most. A traveller may have the information but still need to organize it into a realistic sequence. This manual organization can be time-consuming, particularly when the group has multiple interests or a fixed budget.

2. Primary Problem Statement

Travellers need a simple way to turn their destination, dates, budget, number of travellers and interests into a structured itinerary that can be reviewed, saved and reused, without manually creating every day of the plan from scattered sources.

3. Related Problems

Scattered planning inputs

Travel preferences may be kept in notes or messages. Collecting them in one form reduces the effort required to start planning and gives the AI service a consistent set of inputs.

Unstructured recommendations

A list of attractions does not tell the traveller how to distribute activities across the trip. A day-wise structure with morning, afternoon and evening activities makes the output easier to review and adjust.

Difficulty returning to an earlier plan

A generated itinerary is more useful when the user can revisit it. Saved Trips and database persistence address this need by allowing the user to list and open stored trips.

Finding nearby services

A plan often leads to a separate search for hotels and restaurants. The current application provides destination-based Google Maps search links to help continue that research externally.

4. Objectives

1. Collect trip preferences in a single interface.
2. Generate a structured itinerary using a hosted generative AI model.
3. Validate the AI response before returning it to the frontend.
4. Persist trip and itinerary records in Supabase.
5. Allow saved trips to be listed, retrieved and deleted.
6. Provide hotel and restaurant search links with restaurant food-preference terms.
7. Allow the itinerary to be printed or saved as a PDF through the browser.

5. Expected Outcomes and Validation

The project should be judged by the behaviour of the complete workflow rather than by the presence of individual screens alone. A valid request should return a structured plan; the user should be able to review the result, save it, retrieve it later and delete it. Map searches should open an external search query. The print action should invoke the browser print interface.

Formal performance results, user interviews and usability scores were not included in the supplied source materials. They should only be added if actually collected. Likewise, generated costs are estimates and should not be described as verified prices.